

OVERVIEW OF PHYSICAL AND DATA STORAGE STRUCTURE

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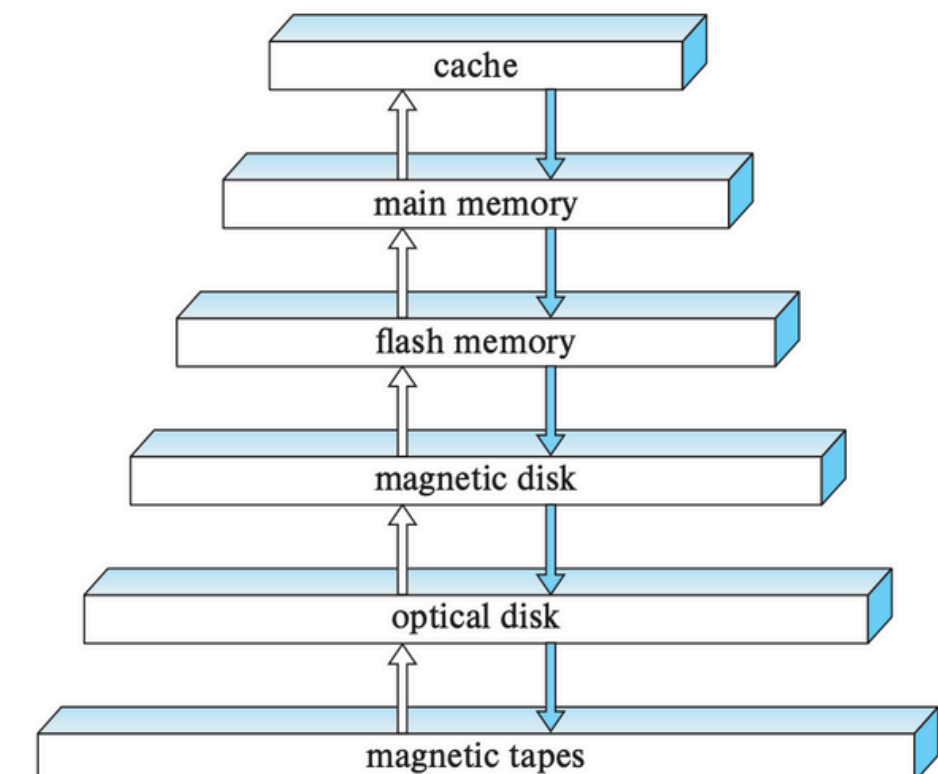
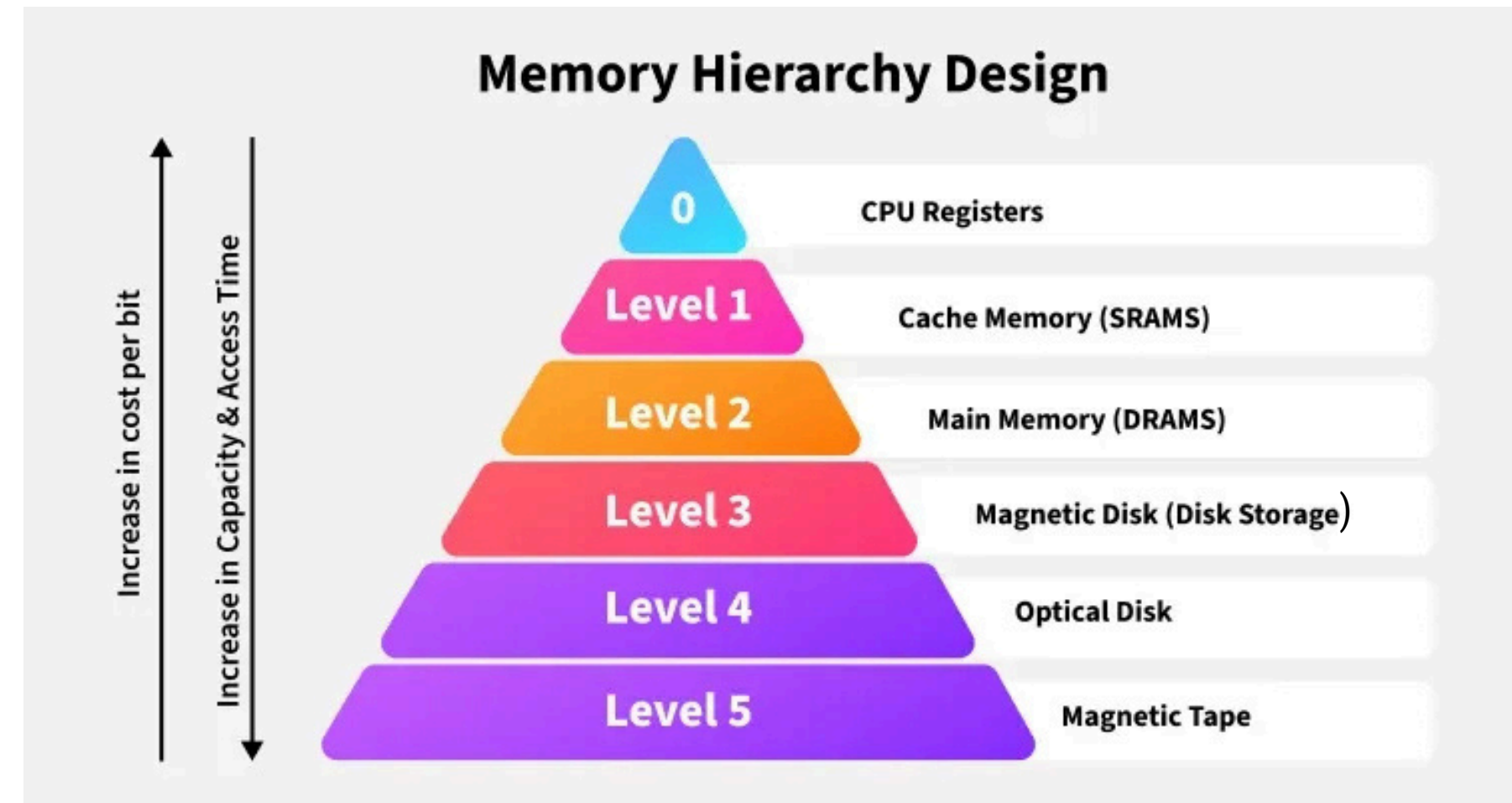


INTRODUCTION TO PHYSICAL STORAGE SYSTEMS

Definition: Low-level management of data storage to optimize access speed and ensure durability.

- *Efficient Data Storage: Organize and store data in a manner that optimizes space and retrieval processes.*
- *Durability: Ensure long-term reliability of stored data, even in the event of power failures or system crashes.*
- *Fast Access: Enable rapid data retrieval for applications requiring high-performance computing or real-time analytics*

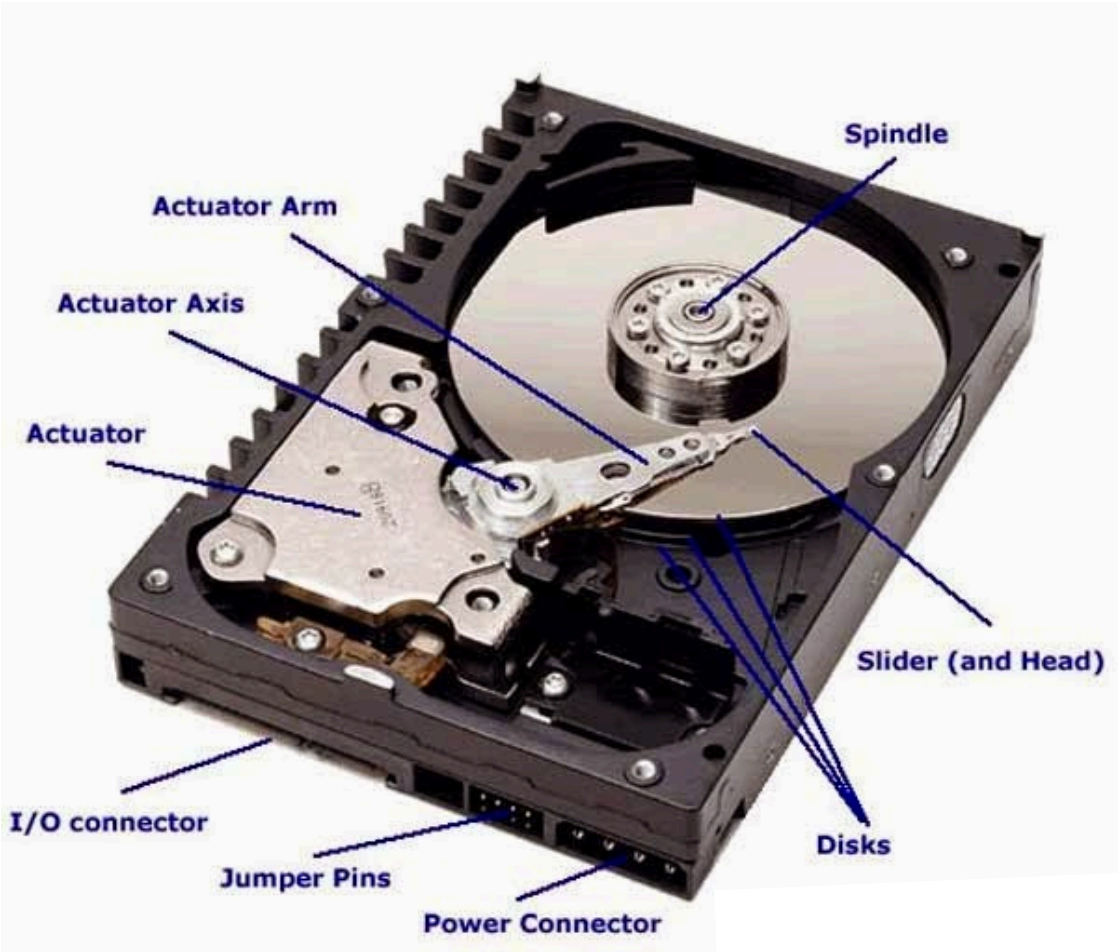
Eg-Hard Disks, SSDs, Magnetic Tapes etc



TYPES OF STORAGE MEDIA

Comparison of SSD vs HDD Performance

Feature	SSD	HDD
Speed	500–3,500+ MB/s	30–150 MB/s
Cost	Higher per GB	Lower per GB
Durability	No moving parts, highly durable	Mechanical parts prone to wear
Power Efficiency	More energy-efficient	Higher power consumption
Noise	Silent	Audible due to spinning disks 2 6 8 .



MAGNETIC DISC

Cache

- Fastest, closest to the CPU, stores temporary data. Small and expensive, used to reduce latency.

Main Memory (RAM)

- High-speed, volatile memory for active processes. More affordable than cache but still costly.

Flash Memory

- Non-volatile, faster than HDDs, moderate cost. Commonly used in SSDs and USB drives.

Magnetic Disk (HDD)

- Low-cost, high-capacity storage with slower speeds due to mechanical parts.

Optical & Tape Storage

- Inexpensive, durable, sequential-access storage for backups and archiving.

STORAGE INTERFACES

Direct Disk Interfaces

- SATA: Common interface for HDDs and SSDs, supports up to 6 Gbps (600 MB/s). Cost-effective but slower compared to SAS and NVMe.
- SAS: Enterprise-grade interface with speeds up to 12 Gbps. Offers reliability features like hot-swapping and redundancy, mainly used in servers.
- NVMe: Optimized for SSDs, uses the PCIe bus for speeds up to 8 GB/s (x4 lanes). Provides low latency and high scalability, ideal for high-performance applications.

Networked Storage

- SAN: Connects multiple disks via high-speed networks using protocols like iSCSI, Fiber Channel, or InfiniBand. Offers a logical view of large, reliable storage arrays with RAID.
- NAS: Provides file system access over networks using protocols like NFS or CIFS. Suitable for shared storage environments.

Cloud Storage

- Data stored remotely and accessed via APIs. High latency makes it less suitable for databases but ideal for object storage in applications

RAID (REDUNDANT ARRAYS OF INDEPENDENT DISKS)

RAID combines multiple disks to improve performance and reliability. It is widely used in applications like web, database, and multimedia systems where large-scale data storage and high availability are critical.

Redundancy for Reliability

- **Mirroring:** A technique where data is duplicated across two disks. If one disk fails, the other can recover the data. Mirrored systems offer high reliability with mean time to data loss ranging from 55 to 110 years but are expensive due to duplicate storage requirements.
- **Parity:** Parity blocks are used in RAID levels like RAID 5 and RAID 6 to ensure data recovery without full duplication. Parity allows reconstruction of lost data by using XOR operations on remaining blocks and parity blocks.

Parallelism for Performance

- **Striping:** Data is split across multiple disks (e.g., block-level striping). This improves performance by enabling simultaneous reads/writes across disks, increasing throughput for large file transfers.
- **Block-Level Striping:** Logical blocks are distributed across disks in a round-robin manner (e.g., disk $(i \bmod n) + 1$). This allows faster large-file reads while freeing other disks for simultaneous operations.

RAID LEVELS

RAID Levels - Key Points

1. RAID 0 (Striping):

- Data is striped across multiple disks to improve performance.
- No redundancy, meaning failure of one disk results in data loss.

2. RAID 1 (Mirroring):

- Data is duplicated across two disks for high reliability.
- Expensive due to duplicate storage but ensures data recovery in case of disk failure.

3. RAID 5 (Distributed Parity):

- Combines striping with parity blocks distributed across all disks.
- Allows recovery from a single disk failure using XOR operations on parity and remaining data blocks.

4. RAID 6 (Dual Parity):

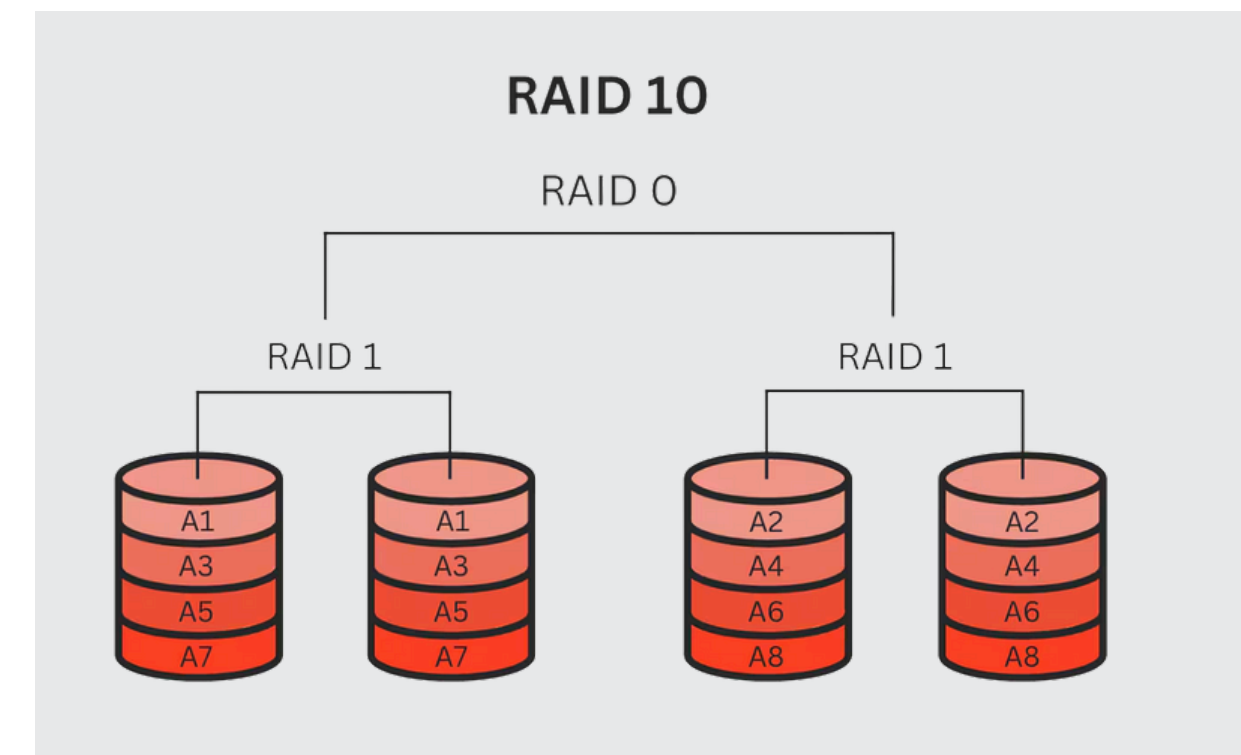
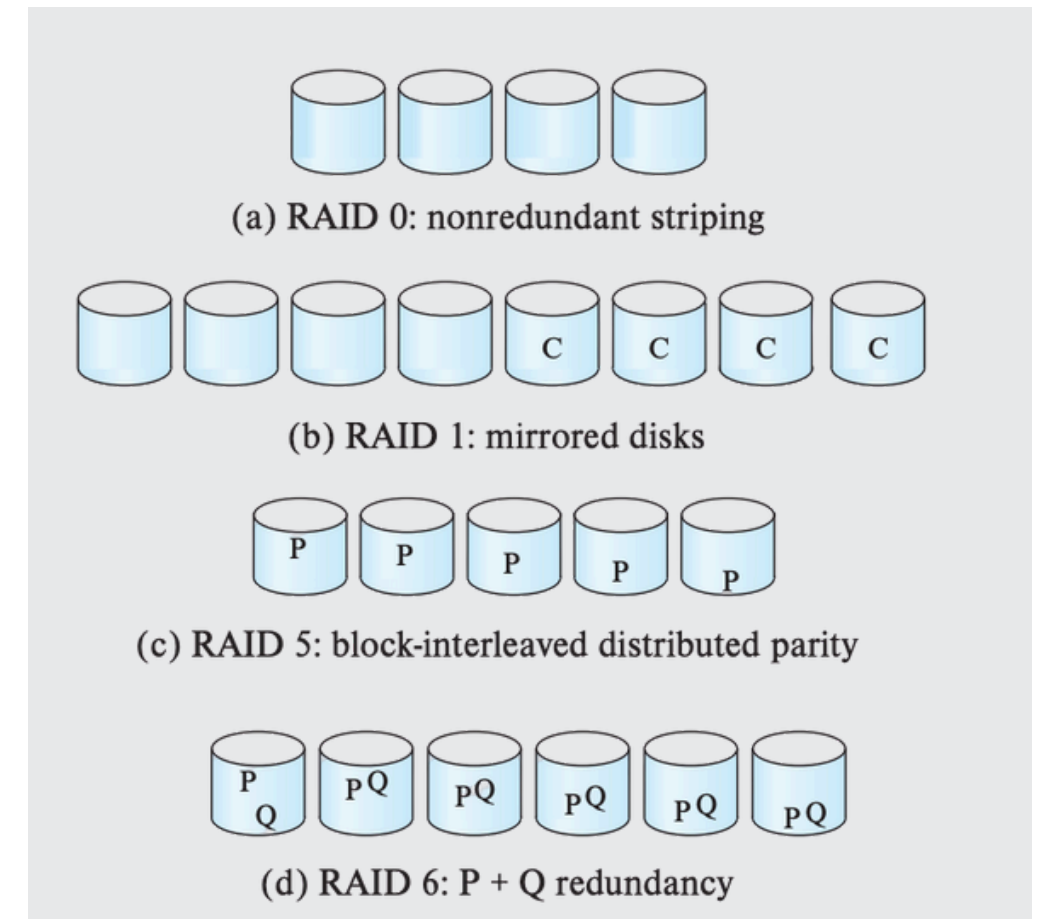
- Similar to RAID 5 but includes additional redundancy using error-correcting codes (e.g., Reed-Solomon).
- Can tolerate up to two disk failures, making it ideal for critical systems.

5. Parity Mechanism:

- Parity blocks store redundant information calculated using XOR operations on data blocks.
- Enables reconstruction of lost data without full duplication, reducing costs compared to mirroring.

6. Nested RAID Levels:

- RAID 10 (1+0): Combines mirroring and striping for optimal performance and reliability.
- Used in scenarios requiring both high speed and fault tolerance.



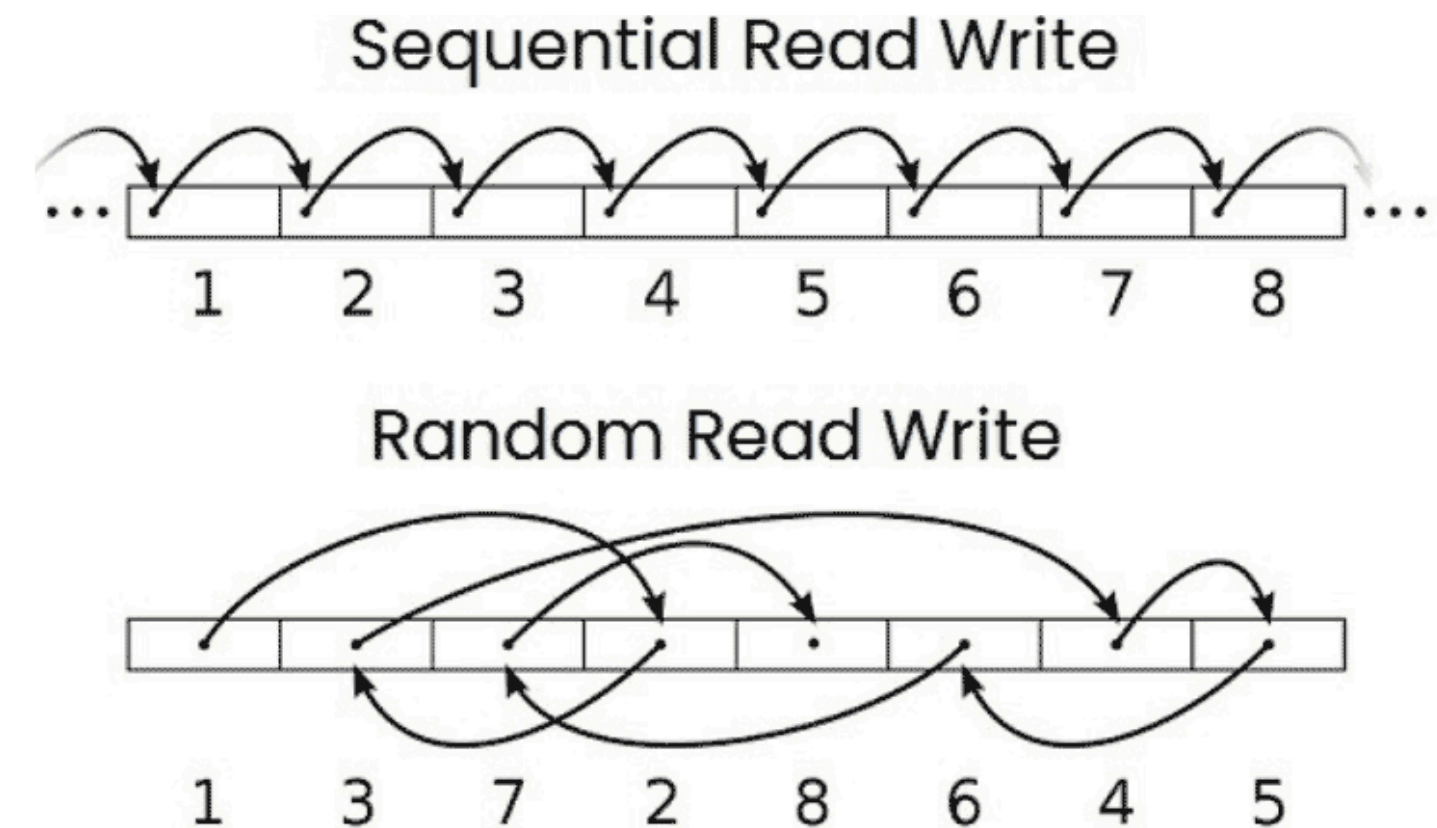
DISK-BLOCK ACCESS

Overview:

1. Requests for disk I/O are generated by the query processing subsystem.
2. Data is transferred between disk and main memory in units of blocks.
3. Requests specify:
 - Disk Identifier & Logical Block Number (Direct Disk Access).
 - File Identifier & Block Number (Operating System Files).

Access Patterns:

- Sequential Access: Successive blocks from the same or adjacent tracks (Fast).
- Random Access: Blocks located randomly on disk, requiring seeks (Slow).



TECHNIQUES TO IMPROVE DISK-BLOCK ACCESS

- **Buffering:**
Temporarily stores recently accessed blocks in main memory for quicker future access.
Performed by both the operating system and the database management system.
- **Read-Ahead:**
Continuously reads consecutive disk blocks into memory even when not immediately requested.
Highly effective for sequential access patterns, but not useful for random access.
- **Scheduling (Elevator Algorithm):**
Efficiently orders disk access requests to reduce overall disk-arm movement.
Moves in one direction servicing all requests until none remain before reversing direction.
- **File Organization & Fragmentation:**
Storing data blocks in a way that closely matches the expected access patterns.
Fragmentation can occur due to multiple small appends, leading to scattered blocks.
Defragmentation tools can reorganize blocks to restore them contiguously on disk.
- **Non-Volatile Write Buffers (NVRAM):**
Temporarily stores data in non-volatile memory to prevent data loss during power failure.
Provides a significant speedup for write-intensive applications by acknowledging writes instantly.